

HiddenPrompt - Product Requirements Document (PRD)

1. Project Overview

HiddenPrompt is a real-time multiplayer AI guessing game where players race to identify a secret prompt from AI-generated surreal images. Instead of humans drawing, AI becomes the artist, making the game accessible and fun regardless of drawing skill.

2. Problem Statement

Traditional drawing games rely on artistic ability, which can disadvantage many players. HiddenPrompt removes that barrier by using AI-generated abstract artwork while preserving the excitement of guessing.

3. Goals

- Create multiplayer rooms without login.
- Join using only a username and room code.
- Play multiple rounds.
- Guess AI-generated images in real time.
- Award points based on speed.
- Declare an overall winner.

4. Target Users

Friends, college students, families, Discord communities, and streamers looking for a casual multiplayer party game.

5. Core Features

- Create Room
- Join Room
- Lobby with ready status
- Host-controlled settings
- AI image generation
- Real-time guessing chat
- Dynamic AI hint regeneration
- Live leaderboard

6. User Flow

Landing Page → Create/Join Room → Enter Username → Lobby → Host Starts Game → Prompt Selection → AI Generates Image → Players Guess → Scoreboard → Next Round → Final Winner

7. Functional Requirements

Lobby: Host creates a room, players join via room code, host configures rounds, timer, difficulty, and starts the game.

Prompt Selection: The active prompter chooses one prompt from three predefined options.

AI Generation: Backend generates an abstract image. If nobody guesses within the hint interval, a new interpretation of the same prompt is generated.

Guessing: Unlimited guesses with spam protection (1 guess/sec). Correct guesses are case-insensitive and each player can score only once per round.

Scoring: Faster correct guesses receive more points. The prompter receives bonus points only if at least one player guesses correctly.

Round End: Ends when timer expires or all eligible players have guessed correctly.

8. Non-Functional Requirements

- Real-time synchronization using WebSockets.
- Responsive UI.
- Reconnection support.
- Low latency gameplay.
- No authentication required for MVP.

9. Edge Cases

- Host disconnects → Transfer host.
- Player disconnects → Allow reconnect if room is active.
- AI generation fails → Retry once, then choose a new prompt.
- Empty room → Delete after timeout.

10. Success Criteria

- Players can create and join rooms.
- AI images generate successfully.
- Scores update in real time.
- Winner is calculated correctly.
- Gameplay is synchronized across all connected players.

11. MVP Scope

Store all game state in server memory (RAM). MongoDB, authentication, persistent leaderboards, and match history will be added in future versions.